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Functional Requirements

Functional requirements describe the features and operations that the Flood Prediction System should perform.

1. User Input Management

The system should allow users to enter flood-related environmental parameters such as:

- Latitude
- Longitude
- Rainfall
- Elevation
- Slope
- Distance from water bodies
- Flood duration
- Other geographical factors

2. Data Processing

The system should process the user-provided input data and perform necessary preprocessing before sending it to the machine learning model.

The preprocessing steps include:

- Data cleaning
- Feature selection
- Data transformation
- Handling missing values

3. Machine Learning Prediction

The system should use a trained machine learning model to analyze input parameters and

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predict flood risk.

The model should:

- Load the trained model
- Accept user input
- Generate flood predictions
- Classify risk levels

4. Result Display

The system should display the prediction result through a user-friendly web interface.

The output should include:

- Flood risk level
- Prediction status
- Relevant information for users

5. Model Management

The system should allow developers to:

- Train the machine learning model
- Evaluate model performance
- Update the model with new data

6. Web Application

The system should provide an interactive Flask-based web application where users can easily enter details and view predictions.